

BHARATH P

Software Engineer | Full Stack LLM Developer | Python, RAG & Agentic AI

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PROFESSIONAL SUMMARY

Computer Science graduate (9.0 CGPA, REVA University) specializing in Python, Machine Learning, and full-stack LLM application development. Delivered a production-grade AI-powered Slack bot for Red Hat — integrating Google Gemini, real-time shift scheduling, and multi-turn conversation flows. Hands-on experience building RAG pipelines, LangChain/LangGraph agents, and deploying ML models via Flask/Streamlit. Seeking full-time Software Engineer / AI Engineer roles where I can own projects end-to-end.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript (basics)

AI / ML / LLM: LangChain, LangGraph, RAG (hybrid, HyDE, Self-RAG), Groq, HuggingFace, scikit-learn, Chroma/pgvector, Tavily, tool-calling agents

Web / Backend: Flask, FastAPI (learning), Streamlit, REST APIs

Tools / Platforms: Slack Bolt SDK, Git/GitHub, Brevo (email API), Docker (learning), Streamlit Cloud, Render

PROFESSIONAL EXPERIENCE

Python & AI/ML Instructor — Aptech Computer Education 2 Years

Hesaraghatta, Bengaluru | Python, Machine Learning, AI

- Awarded for Best Teaching in Coding, recognized for instructional quality and student outcomes in Python and AI/ML courses.
- Taught Python programming, Machine Learning, and applied AI to student cohorts, mentoring learners from fundamentals through project-level work.
- Managed the institution's social media, advertising, and Instagram presence alongside instructional duties.

PROJECTS

RedHatDocs AI Support Bot2024 – 2025

Client: Red Hat (Internal Tools) | Python, Slack Bolt, Google Gemini 2.5 Flash, Socket Mode, Threading

- Built and deployed a production Slack bot for Red Hat's internal tech-info workspace, handling LMS, EBS, and Salesforce support queries autonomously.
- Integrated Google Gemini 2.5 Flash API for multi-turn, context-aware Q&A with automated retry and exponential back-off — zero downtime in production.
- Designed a state-machine conversation engine with per-user memory, thread-safe session management, and IST-aware on-call shift scheduling with delayed support pings via background threading.

Full AI Agent from Scratch — Tool Calling & MemoryPersonal Project

Python, Groq API, Streamlit | GitHub | Live Demo

- Built a multi-step tool-calling agent loop from scratch (no framework), enabling the model to autonomously call web search, weather, and memory-read/write functions.
- Implemented file-based persistent memory (user preferences + recent conversations) for personalized, context-aware responses across sessions.

RAG ImplementationsOngoing

Python, LangChain, Chroma, HuggingFace

- Implemented hybrid retrieval (BM25 + Chroma vector search) using LangChain's EnsembleRetriever with Reciprocal Rank Fusion; applied HyDE and Self-RAG across document Q&A projects.

Malicious URL Detector — Random Forest ClassifierPersonal Project

Python, Flask, scikit-learn, pandas | GitHub

- Built a multi-class URL classifier (Benign / Malware / Defacement) using a Random Forest model trained on 12 hand-engineered lexical features.
- Addressed class imbalance using `class_weight="balanced"`; deployed as a Flask app with a whitelist pre-check for trusted domains.

EDUCATION

B.E., Computer Science — REVA University

CGPA: 9.0 / 10